

LearnHub

E-Learning Mobile Application Software Development & Documentation

Tasks 2 — Design Diagrams

7. System Architecture

7.1 Overview

LearnHub follows a layered client-side architecture organized into four distinct layers: Presentation, State Management, Data, and Persistence. This separation of concerns ensures maintainability and scalability.

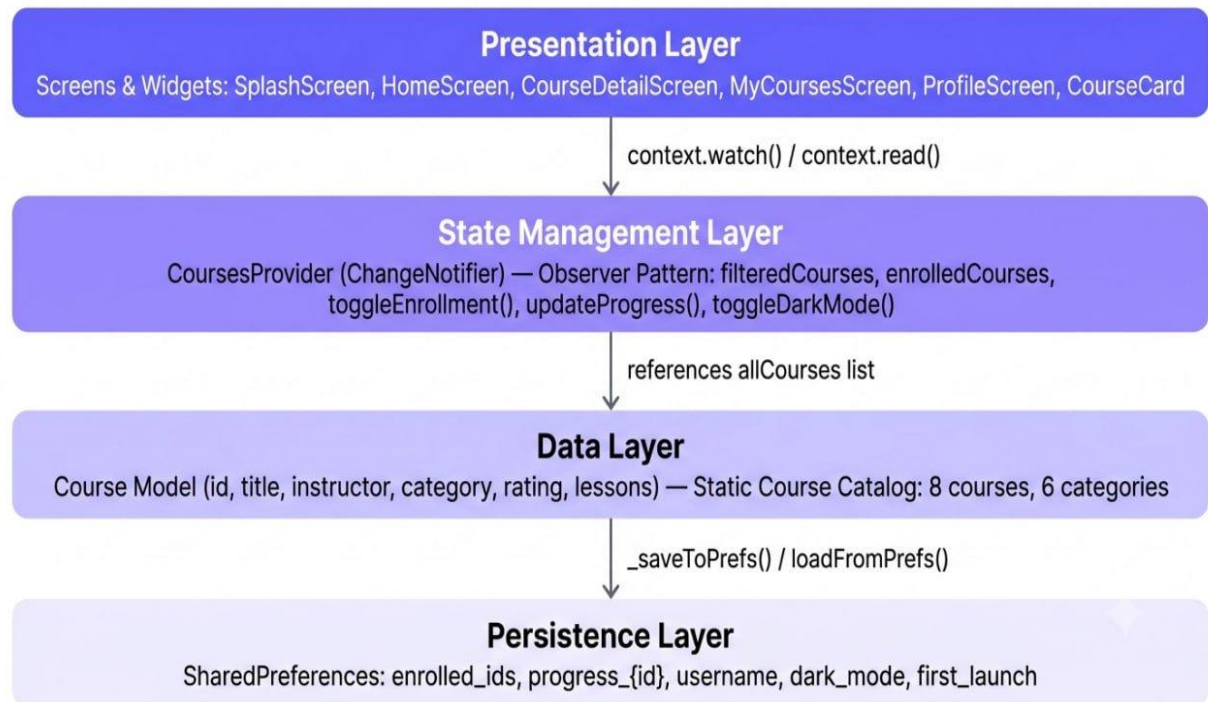
7.2 Architecture Layers

Layer	Components	Responsibility
Presentation Layer	Screens + Widgets	Renders UI and handles user interaction
State Management	CoursesProvider (Provider)	Manages app-wide state via Observer Pattern
Data Layer	Course Model + Static Data	Defines data structures and course catalog
Persistence Layer	SharedPreferences	Stores and retrieves user data locally

7.3 Data Flow

- Screens call `context.watch()` to observe `CoursesProvider` and rebuild on state changes.
- Screens call `context.read()` to trigger actions (enroll, update progress, toggle dark mode).
- `CoursesProvider` references the `allCourses` static list from the Data Layer.
- `CoursesProvider` calls `_saveToPrefs()` after every mutation and `loadFromPrefs()` on startup.

7.4 Architecture Diagram



8. Class Diagram

8.1 Overview

The class diagram shows the main classes in LearnHub, their attributes, methods, and relationships. CoursesProvider is the central class that all screens depend on via the Observer Pattern.

8.2 Key Relationships

- CoursesProvider extends ChangeNotifier (Observer Pattern — Subject).
- All screens depend on CoursesProvider via context.watch() (Observer Pattern — Observers).
- CourseDetailScreen and CourseCard have a composition relationship with Course.
- CoursesProvider has a dependency on Course through the allCourses static list.

8.3 Class Diagram

